

Computer Vision with Deep Learning

Lecture 16: Projective geometry and calibration

slides credit: Ioannis Gkioulekas, Lana Lazebnik, Raymond Yeh, David Lindell, (and whoever mentioned in specific slides)

Announcements

- Project proposal grades are out
- HW3 due April 10
- Midterm grading in progress

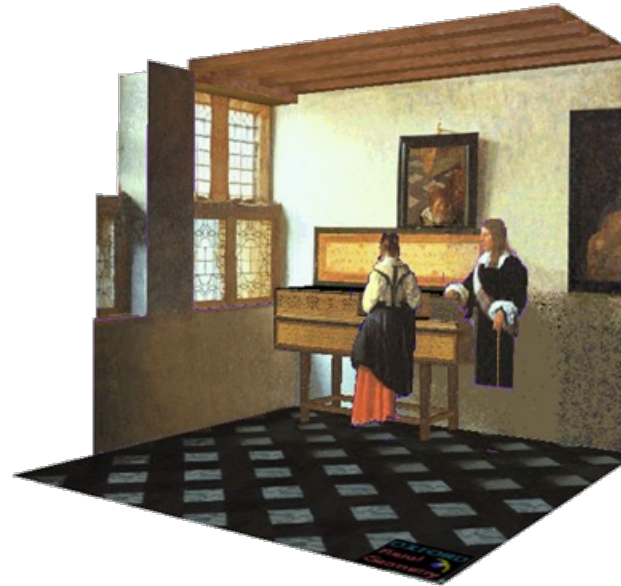
PREVIOUSLY ON

COMPUTER VISION WITH DEEP LEARNING

Our goal: recover 3D information from images

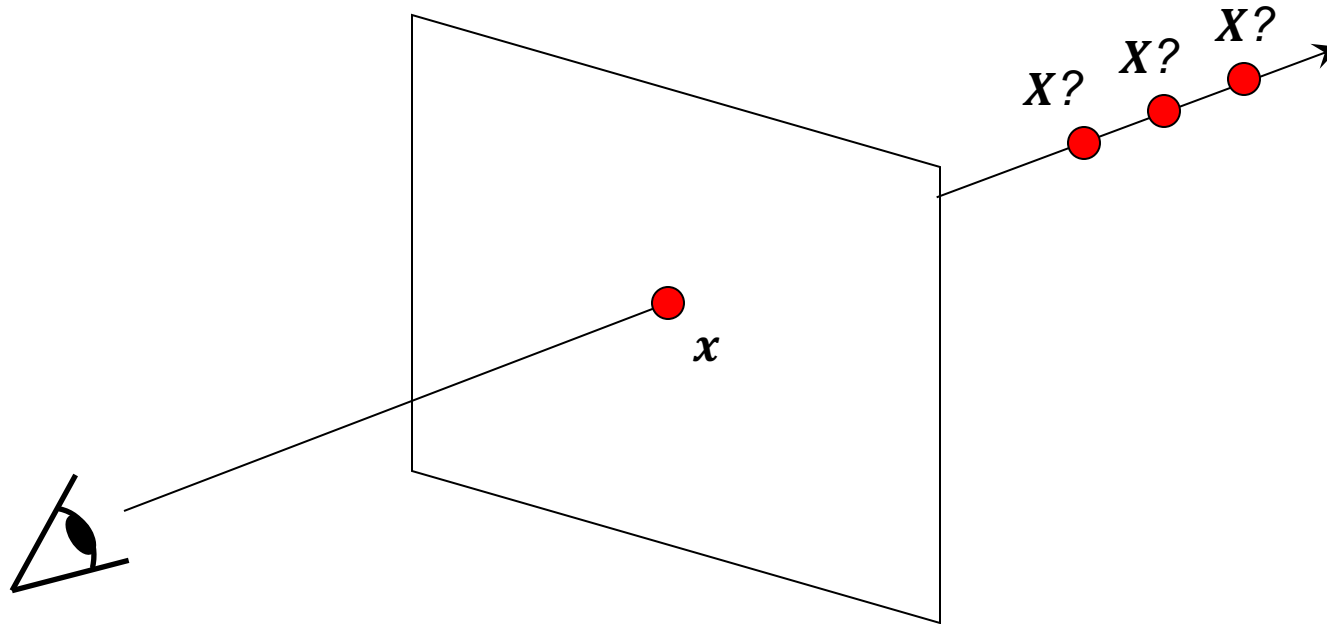


J. Vermeer, Music Lesson, 1662

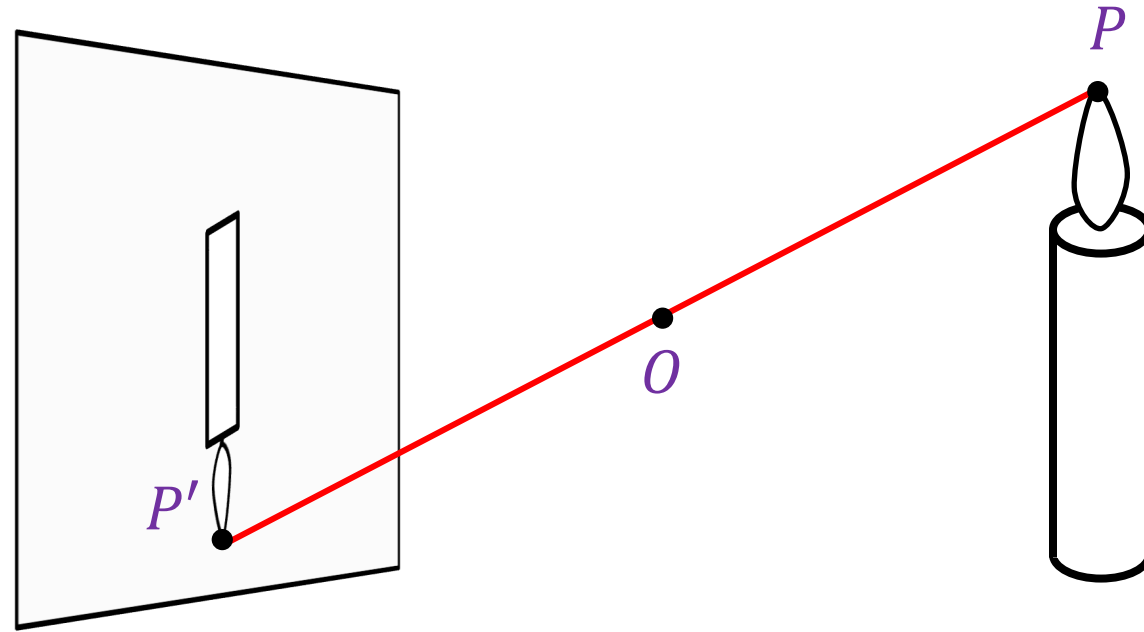


A. Criminisi et al. [Bringing Pictorial Space to Life: computer techniques for the analysis of paintings](#),
Proc. Computers and the History of Art, 2002

Single-view ambiguity

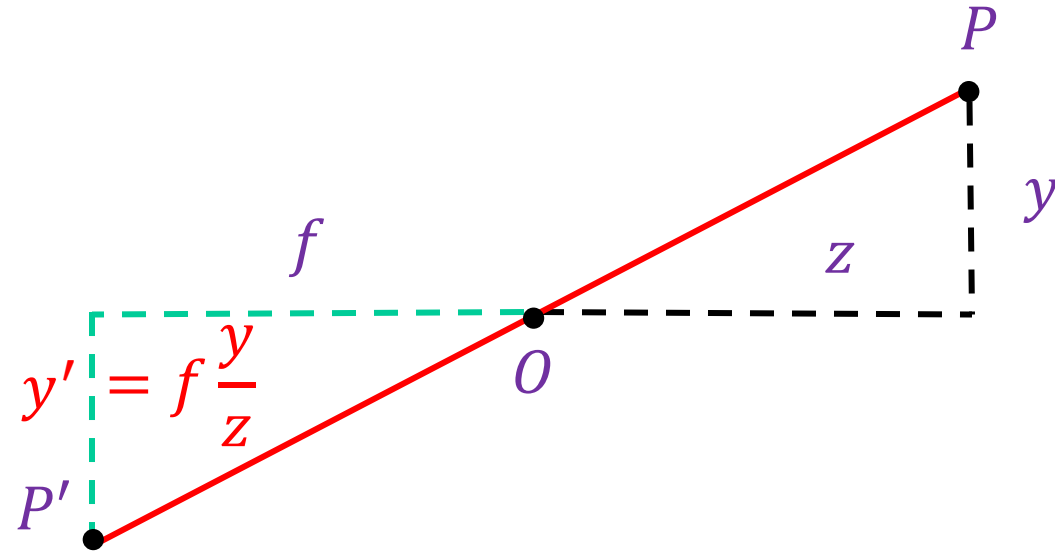


Modeling projection



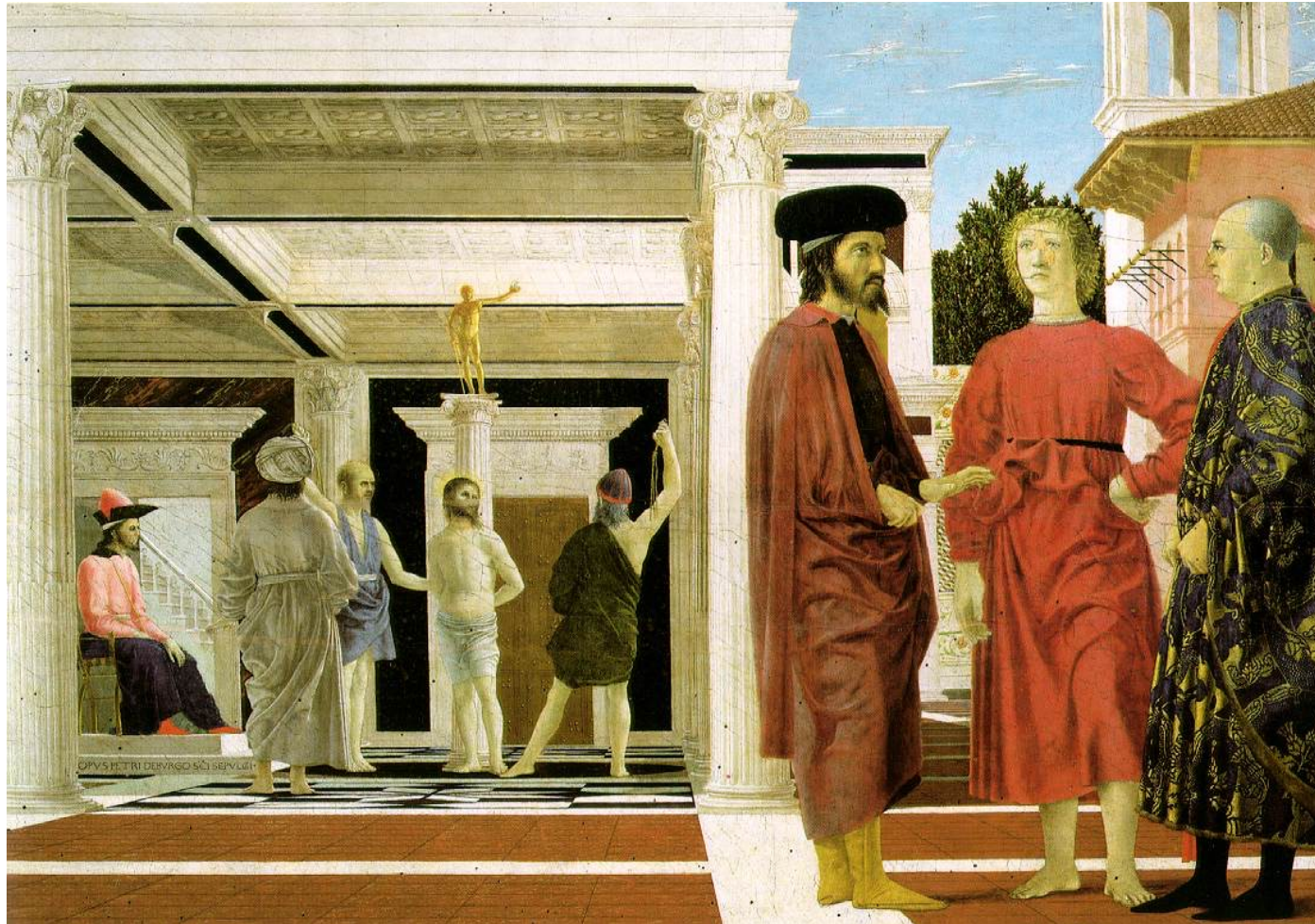
- How do we find the projection P' of a scene point P ?
 - Form the **visual ray** connecting P to the camera center O and find where it intersects the image plane
- All scene points that lie on this visual ray have the same projection in the image
- Are there scene points for which this projection is undefined?

Deriving projected coordinates



$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z} \right)$$

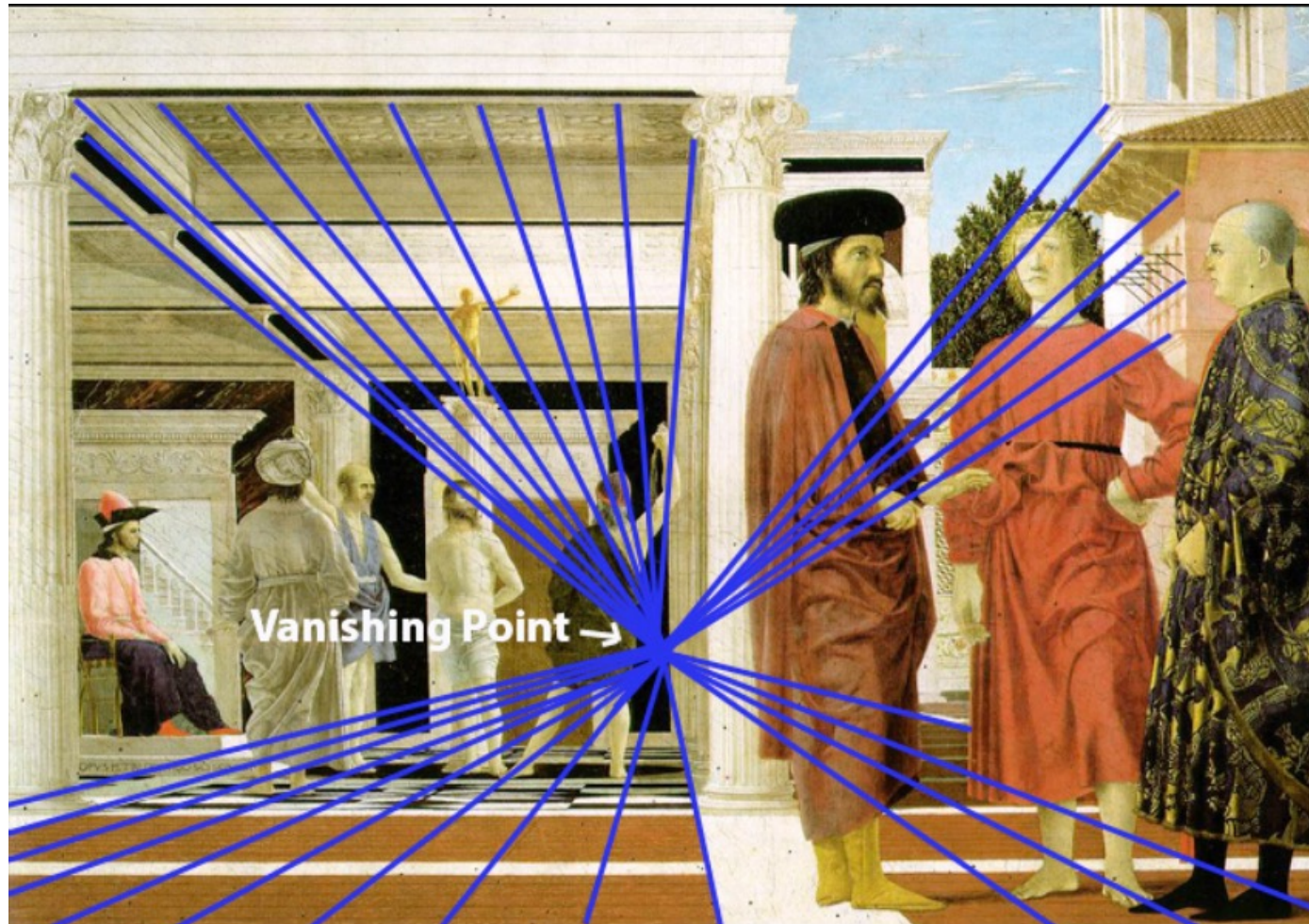
Projection of lines



Piero della Francesca, *Flagellation of Christ*, 1455-1460

Vanishing points

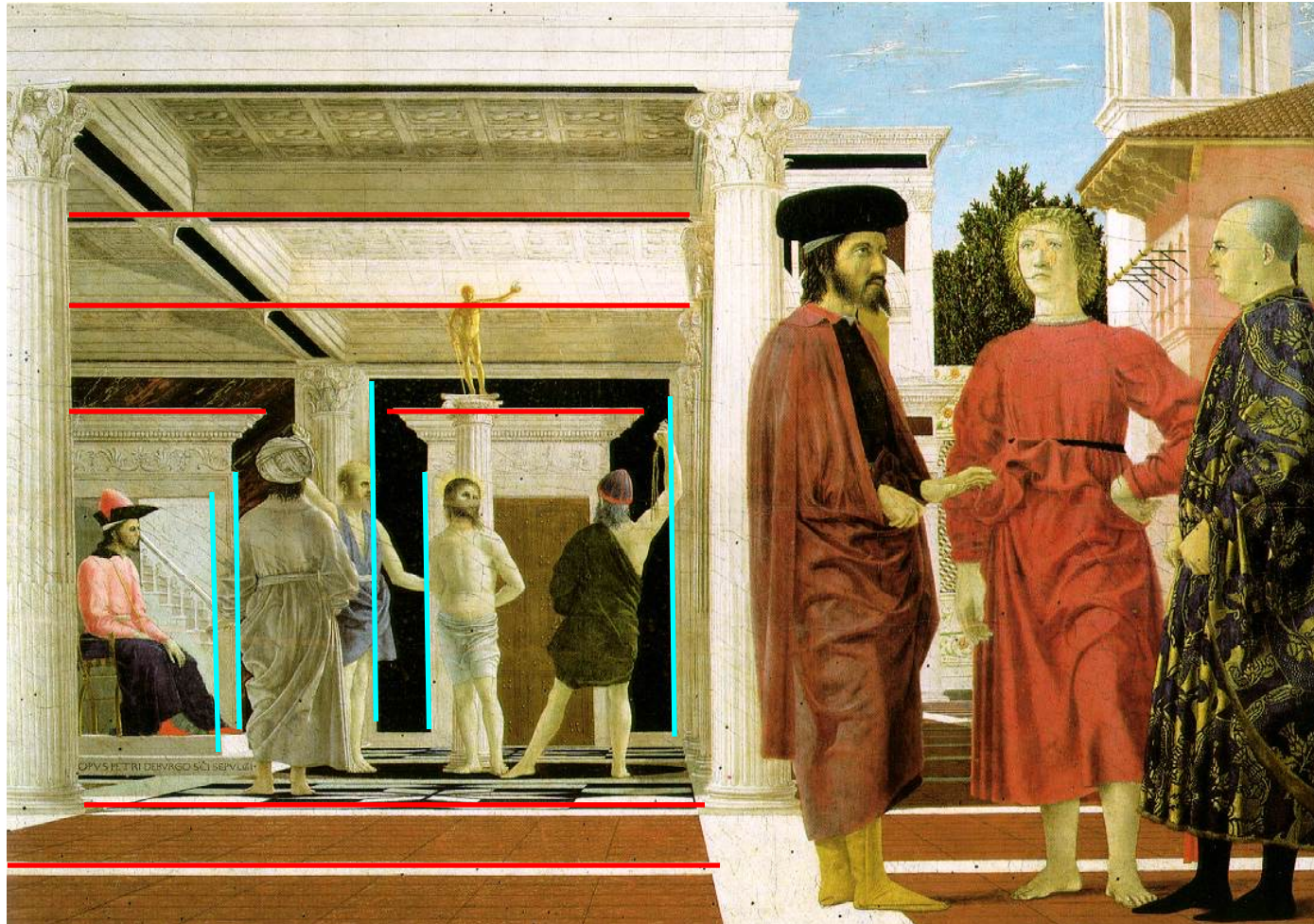
- Parallel lines meet at a *vanishing point*



Piero della Francesca, *Flagellation of Christ*, 1455-1460

Do *all* parallel lines converge to a vanishing point?

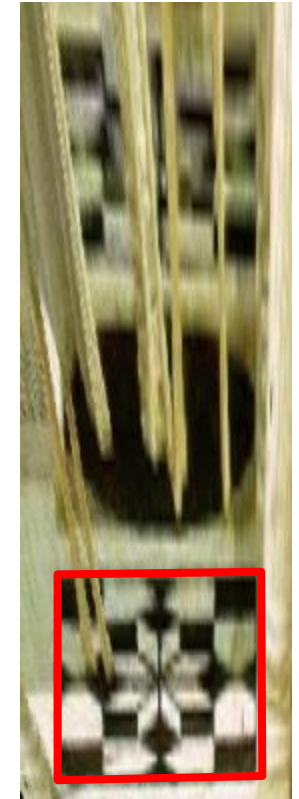
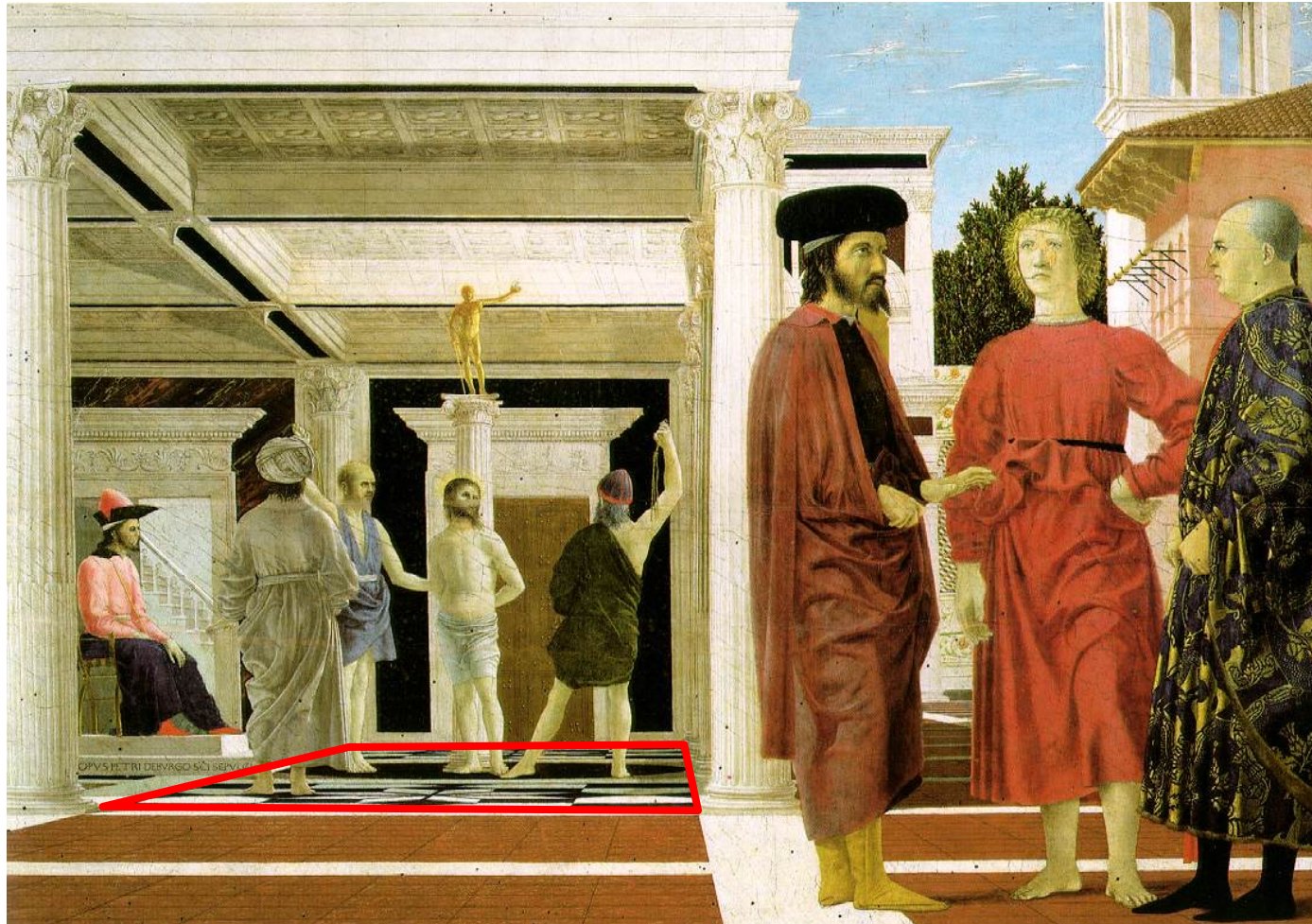
- Not parallel lines that are also parallel to the image plane!



Piero della Francesca, *Flagellation of Christ*, 1455-1460

Projection of planes

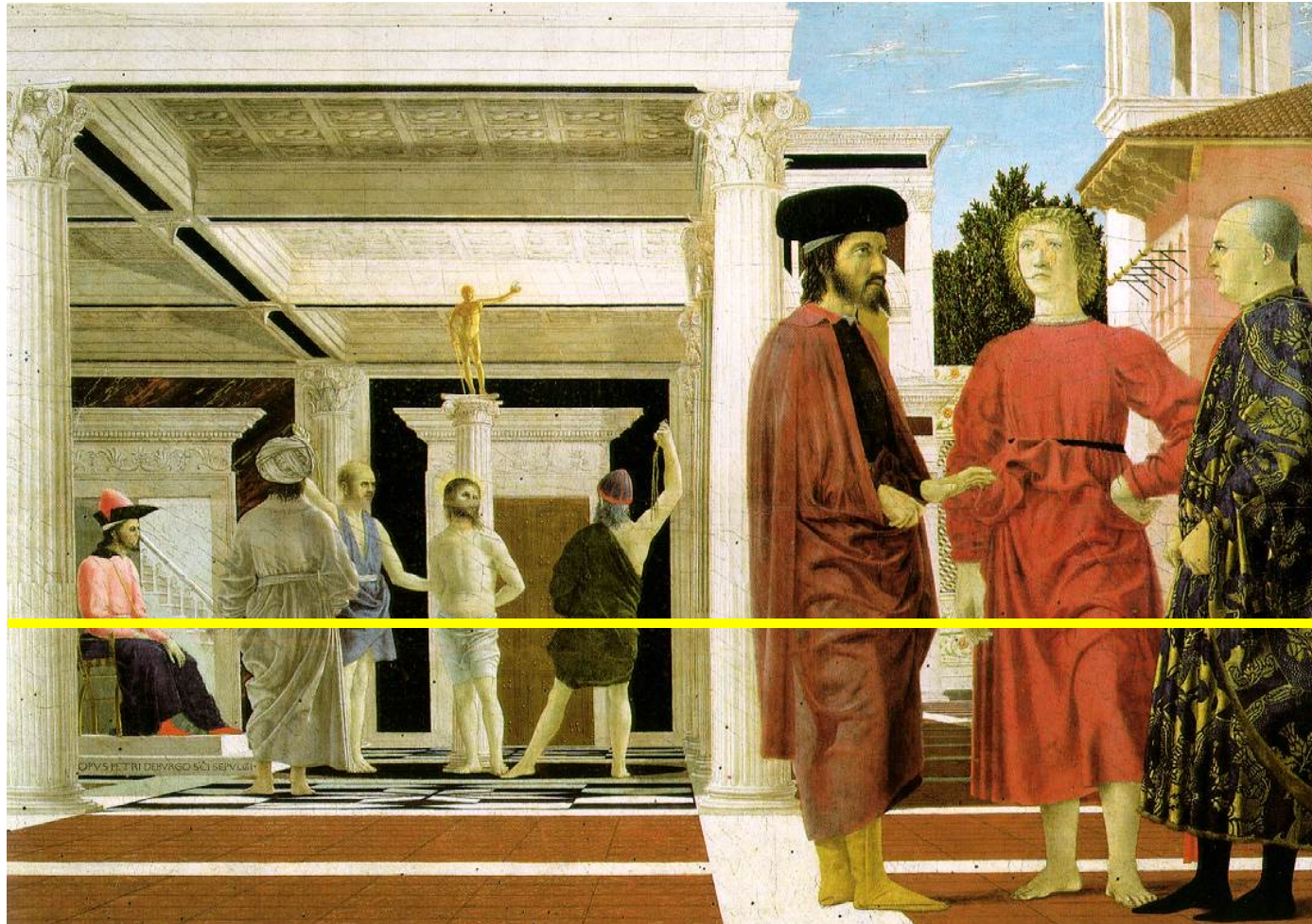
- Patterns on *non-fronto-parallel* planes are distorted by a 2D homography



Piero della Francesca, *Flagellation of Christ*, 1455-1460

Projection of planes

- Non fronto-parallel planes have vanishing lines



Vanishing line of
the ground plane

Piero della Francesca, *Flagellation of Christ*, 1455-1460



horizon = vanishing line



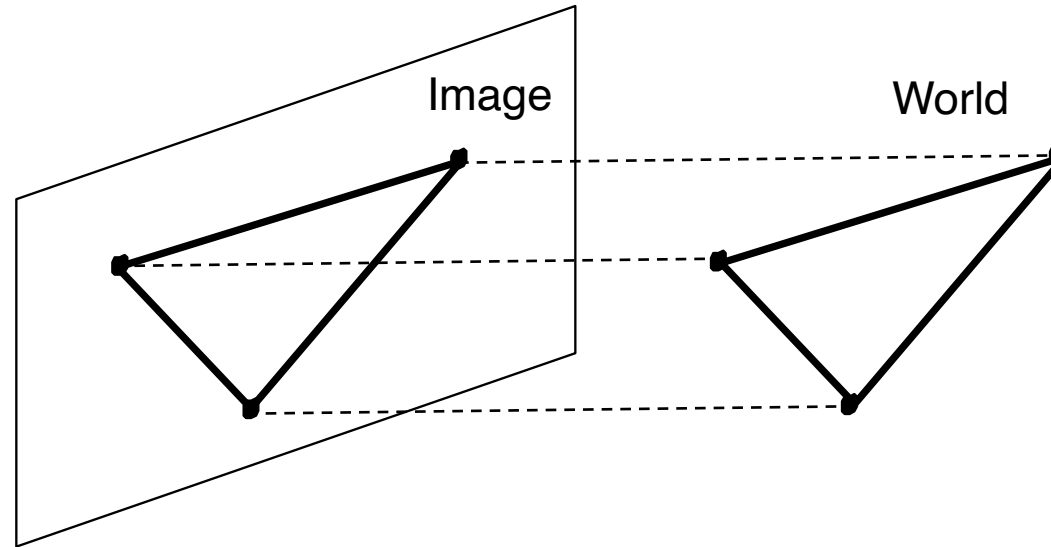
Perspective projection matrix

- Projection is a matrix multiplication using homogeneous coordinates:

$$\begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \\ \\ \end{bmatrix} \Rightarrow \left(f \frac{X}{Z}, f \frac{Y}{Z} \right)$$

divide by the third coordinate

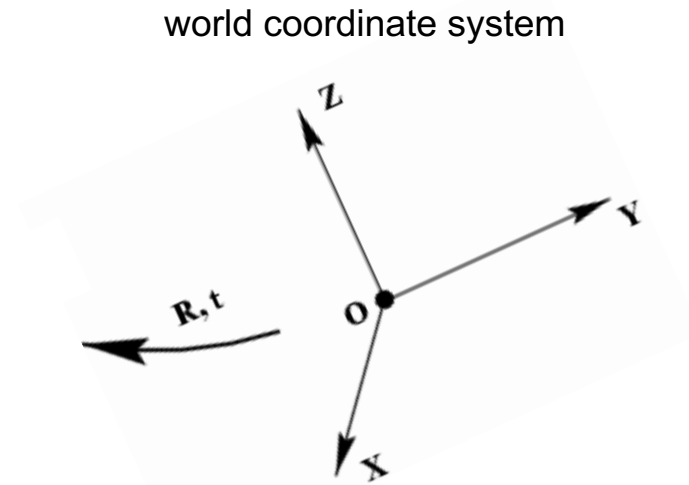
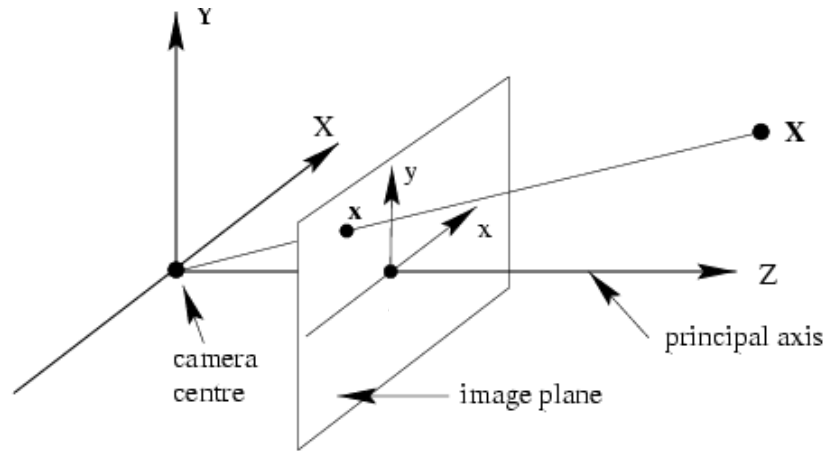
Orthographic projection matrix



- Assuming projection along the z axis, what's the matrix?

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \\ \\ \end{bmatrix}$$

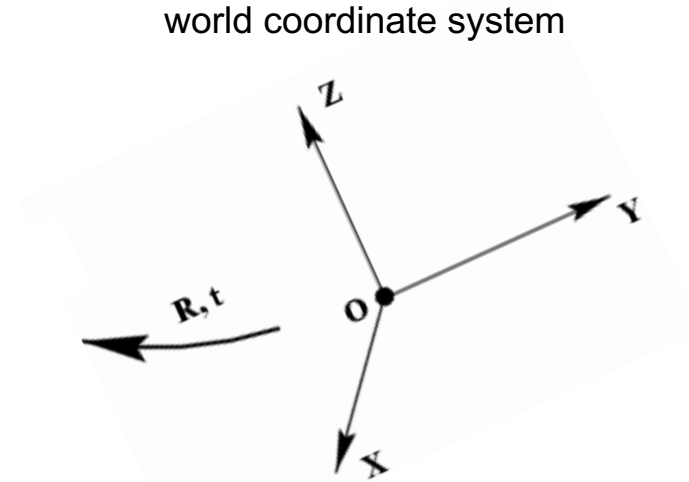
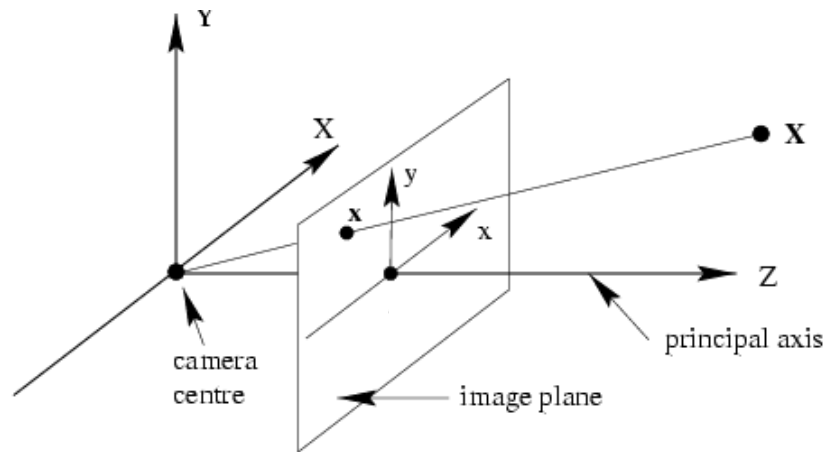
Camera parameters and calibration



- **Camera calibration:** figuring out transformation from *world* coordinate system to *image* coordinate system

$$\begin{array}{c}
 \begin{pmatrix} \text{2D point} \\ (3 \times 1) \end{pmatrix} \\
 \mathbf{x}
 \end{array}
 \cong
 \begin{array}{c}
 \begin{pmatrix} \text{Camera to pixel coord. trans. matrix} \\ \mathbf{K} \ (3 \times 3) \end{pmatrix} \\
 \text{Intrinsic camera parameters: principal point, scaling factors}
 \end{array}
 \begin{array}{c}
 \begin{pmatrix} \text{Canonical projection matrix} \\ \mathbf{[I \mid 0]} \ (3 \times 4) \end{pmatrix} \\
 \text{Extrinsic camera parameters: rotation, translation}
 \end{array}
 \begin{array}{c}
 \begin{pmatrix} \text{World to camera coord. trans. matrix} \\ \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \ (4 \times 4) \end{pmatrix} \\
 \mathbf{X}
 \end{array}
 \begin{array}{c}
 \begin{pmatrix} \text{3D point} \\ (4 \times 1) \end{pmatrix}
 \end{array}$$

Camera parameters and calibration

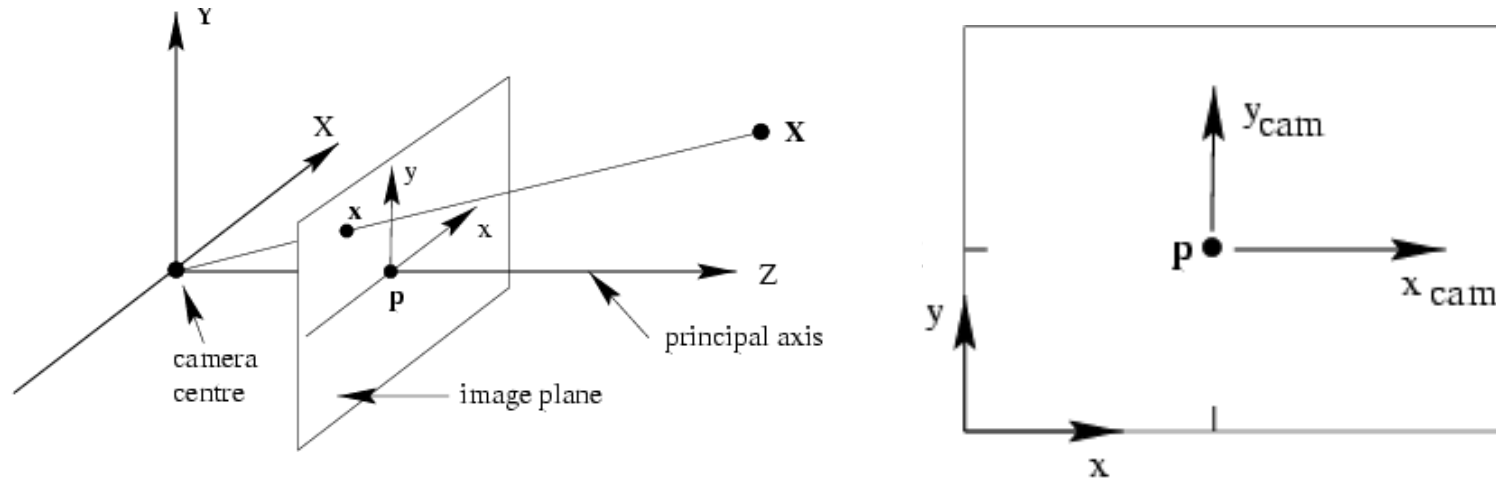


- **Camera calibration:** figuring out transformation from *world* coordinate system to *image* coordinate system

$$\begin{array}{c}
 \begin{pmatrix} \text{2D} \\ \text{point} \\ (3 \times 1) \end{pmatrix} \\
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 \end{array}
 \cong
 \underbrace{
 \begin{pmatrix} \text{Camera to} \\ \text{pixel coord.} \\ \text{trans. matrix} \\ \mathbf{K} \ (3 \times 3) \end{pmatrix}
 \begin{pmatrix} \text{Canonical} \\ \text{projection matrix} \\ [\mathbf{I} \mid \mathbf{0}] \ (3 \times 4) \end{pmatrix}
 \begin{pmatrix} \text{World to} \\ \text{camera coord.} \\ \text{trans. matrix} \\ \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix} \ (4 \times 4) \end{pmatrix}
 }_{\mathbf{P} = \mathbf{K}[\mathbf{R}|\mathbf{t}]}
 \begin{pmatrix} \text{3D} \\ \text{point} \\ (4 \times 1) \end{pmatrix} \\
 \mathbf{X}
 \end{array}$$

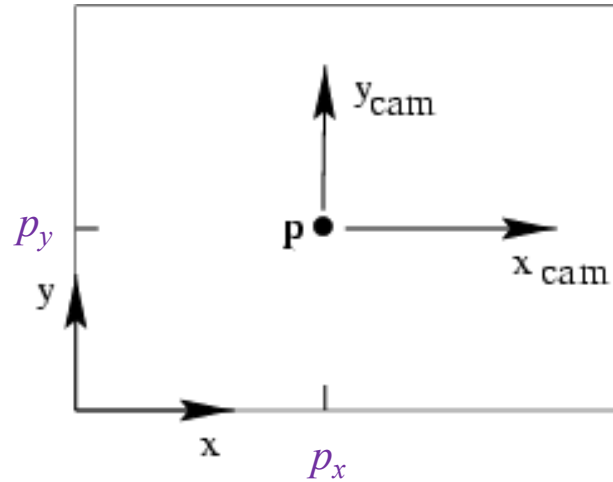
General camera projection matrix

Intrinsic parameters: Principal point



- Principal point (p): point where principal axis intersects the image plane
- In the *normalized* coordinate system, the **origin** of the image is at the **principal point**
- In the *image* coordinate system: the **origin** is in the **corner**

Intrinsic parameters: Principal point

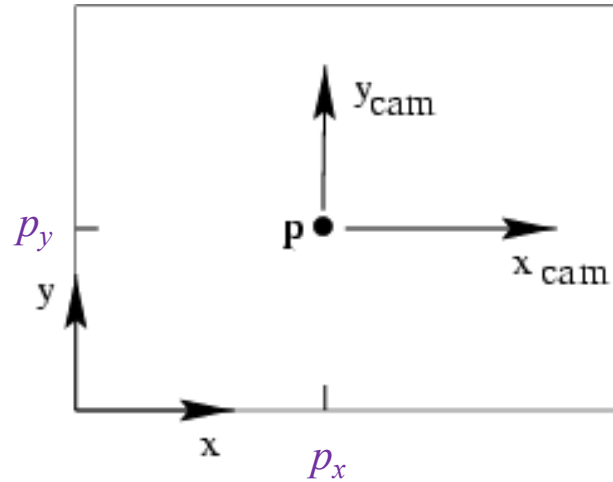


We want the principal point to map to (p_x, p_y) instead of $(0,0)$

$$x = f \frac{X}{Z} + p_x, \quad y = f \frac{Y}{Z} + p_y$$

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \cong \begin{pmatrix} fX + Zp_x \\ fY + Zp_y \\ Z \end{pmatrix} = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Intrinsic parameters: Principal point



Principal point: (p_x, p_y)

$$\begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} f & 0 & p_x & 0 \\ 0 & f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

calibration
matrix $\mathbf{K}$

Canonical
projection matrix
 $[\mathbf{I} \mid \mathbf{0}]$

$$\mathbf{P} = \mathbf{K}[\mathbf{I} \mid \mathbf{0}]$$

Intrinsic parameters: Principal point

- What are the units of the focal length f and principal point coordinates (p_x, p_y) ?
 - Same as world units – presumably metric units
- What units do we want for measuring image coordinates?
 - Pixel units
- Thus, we need to introduce *scaling factors* for mapping from world to pixel units

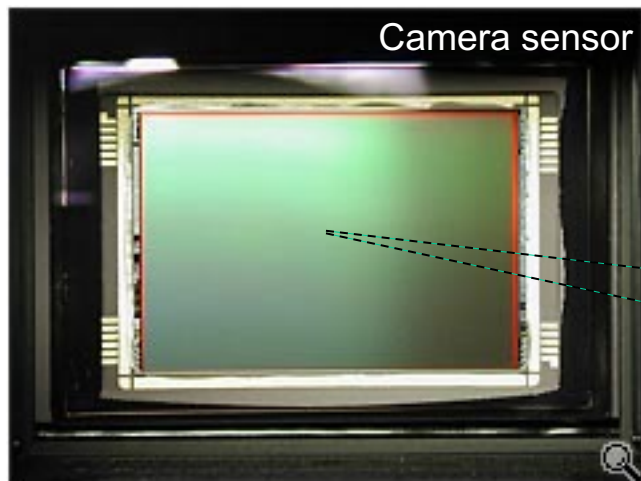
$$\begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} f & 0 & p_x & 0 \\ 0 & f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

calibration
matrix $\mathbf{K}$

Canonical
projection matrix
 $[\mathbf{I} \mid \mathbf{0}]$

$$\mathbf{P} = \mathbf{K}[\mathbf{I} \mid \mathbf{0}]$$

Intrinsic parameters: Scaling factors



Camera sensor

m_x pixels/m in horizontal direction,
 m_y pixels/m in vertical direction

Pixel size (m): $\frac{1}{m_x} \times \frac{1}{m_y}$

Scaling factors

$$\begin{bmatrix} m_x & 0 & 0 \\ 0 & m_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

pixels/m

Calibration matrix
 K in metric units

$$\begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix}$$

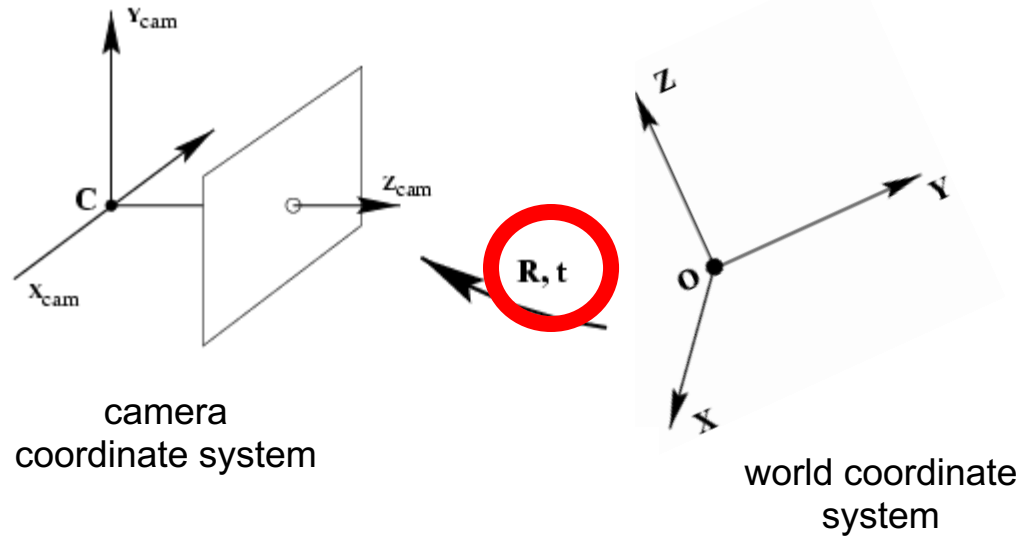
m

Calibration matrix
 K in pixel units

$$\begin{bmatrix} \alpha_x & 0 & \beta_x \\ 0 & \alpha_y & \beta_y \\ 0 & 0 & 1 \end{bmatrix}$$

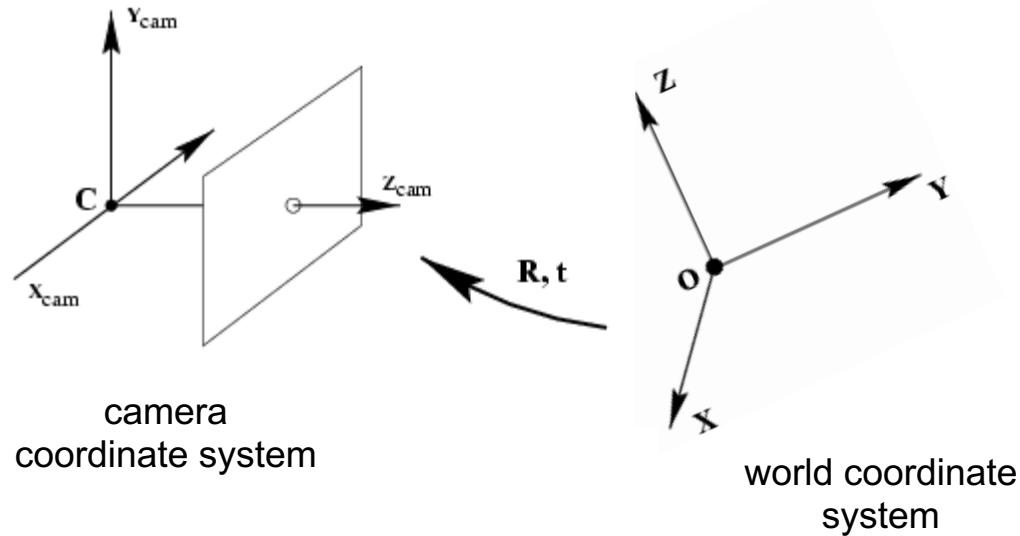
pixels

Extrinsic parameters: Rotation and translation



- In general, the camera coordinate frame will be related to the world coordinate frame by a rotation and a translation

Extrinsic parameters: Rotation and translation



- In general, the camera coordinate frame will be related to the world coordinate frame by a rotation and a translation

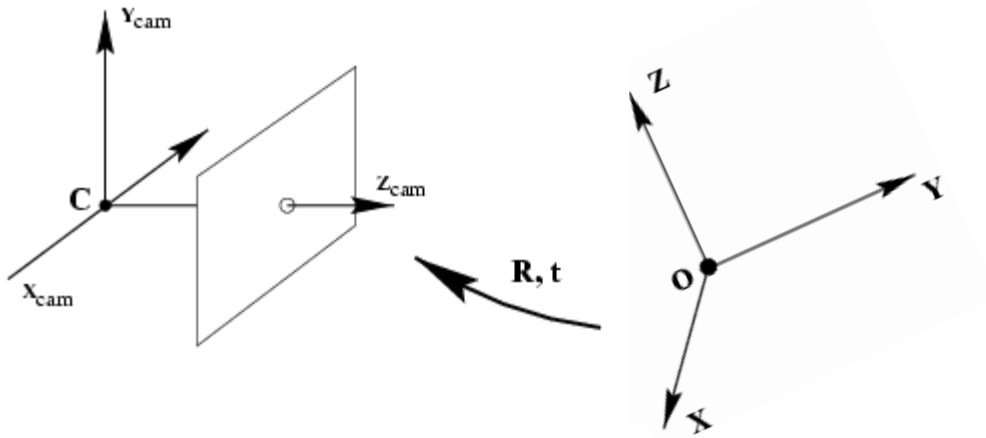
- In non-homogeneous coordinates, the transformation from **world** to normalized **camera** coordinate system is given by:

$$\tilde{X}_{cam} = R(\tilde{X} - \tilde{C}) = R\tilde{X} + t$$

Diagram illustrating the transformation equation:

- $\tilde{X}_{cam}$: coords. of point in normalized camera frame
- R : 3x3 rotation matrix
- $\tilde{X}$: coords. of a point in world frame
- $\tilde{C}$: coords. of camera center in world frame
- t : translation vector

Extrinsic parameters: Rotation and translation



In *non-homogeneous*
coordinates:

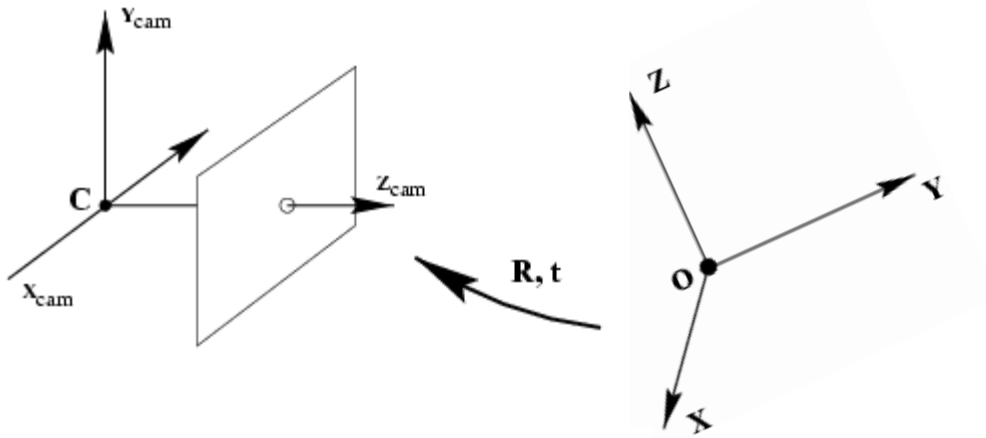
$$\tilde{X}_{\text{cam}} = R\tilde{X} + t$$

In *homogeneous*
coordinates:

$$\begin{pmatrix} \tilde{X}_{\text{cam}} \\ 1 \end{pmatrix} = \begin{bmatrix} R & t \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{pmatrix} \tilde{X} \\ 1 \end{pmatrix}$$

3D transformation
matrix (4 x 4)

Extrinsic parameters: Rotation and translation



In *non-homogeneous* coordinates:

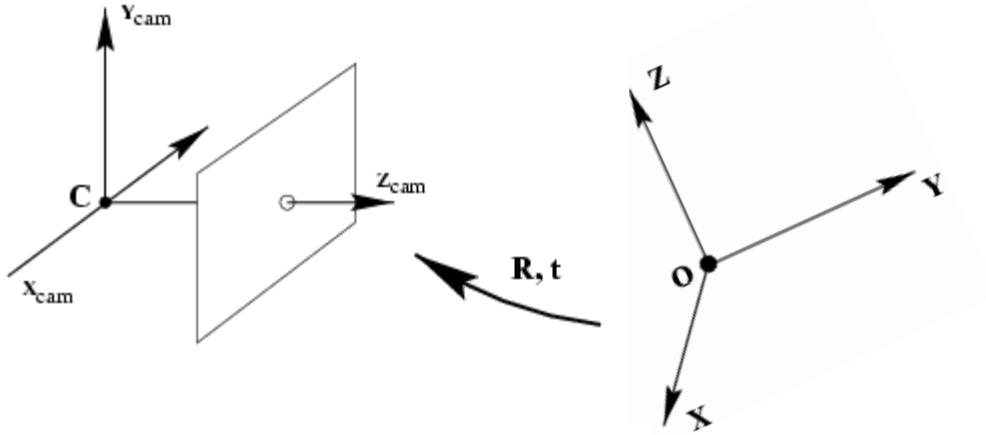
$$\tilde{X}_{cam} = R\tilde{X} + t$$

In *homogeneous* coordinates:

$$X_{cam} = \begin{bmatrix} R & t \\ \mathbf{0}^T & 1 \end{bmatrix} X$$

3D transformation
matrix (4 x 4)

Extrinsic parameters: Rotation and translation



In *non-homogeneous*
coordinates:

$$\tilde{X}_{\text{cam}} = R\tilde{X} + t$$

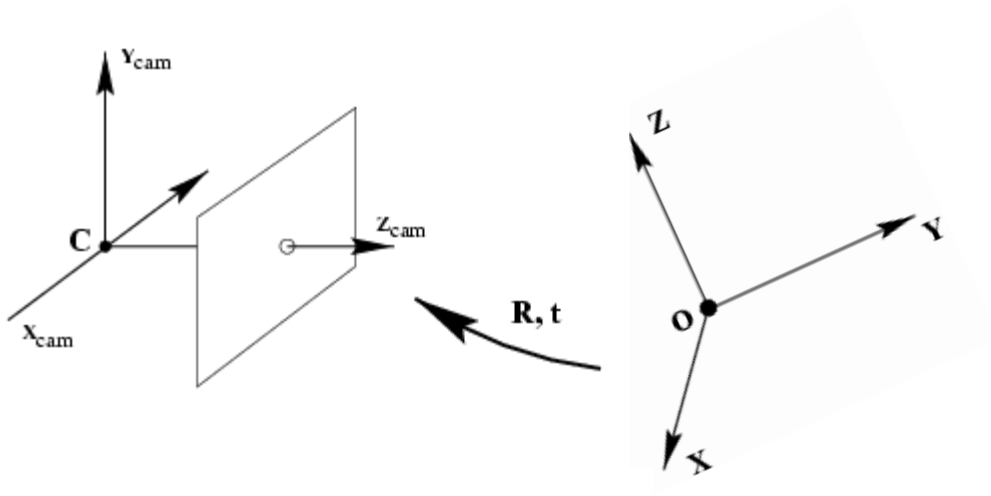
In *homogeneous*
coordinates:

$$\boxed{X_{\text{cam}}} = \begin{bmatrix} R & t \\ \mathbf{0}^T & 1 \end{bmatrix} X$$

Transformation from normalized 3D
coordinates to pixel image coordinates:

$$x \cong K[I|\mathbf{0}]X_{\text{cam}}$$

Extrinsic parameters: Rotation and translation



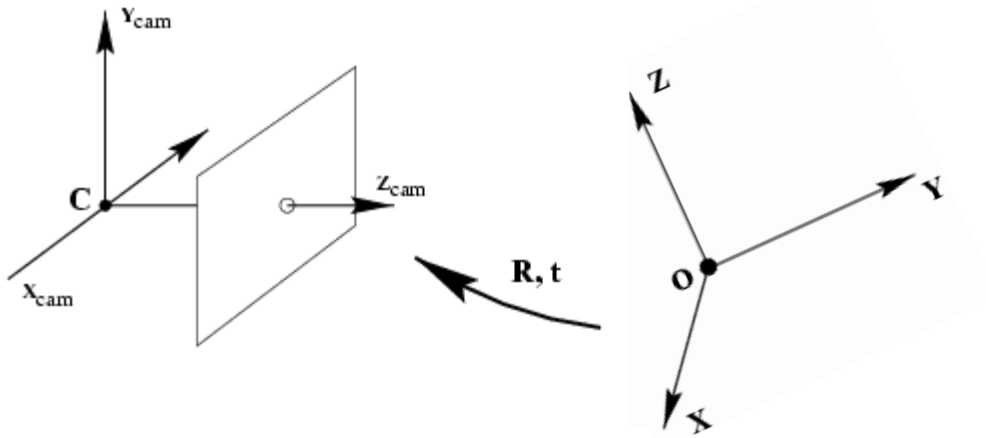
In *homogeneous*
coordinates:

$$x \cong K[I|0] \begin{bmatrix} R & t \\ 0^T & 1 \end{bmatrix} X$$

Finally:

$$x \cong K[R|t]X \quad t = -R\tilde{C}$$

Extrinsic parameters: Rotation and translation



$$x \cong K[R|t]X$$

$$t = -R\tilde{C}$$

coords. of
camera center
in world frame

- What is the projection of the camera center in world coordinates?

$$PC = K[R \mid -R\tilde{C}] \begin{pmatrix} \tilde{C} \\ 1 \end{pmatrix} = K(R\tilde{C} - R\tilde{C}) = 0$$

- The camera center is the **null space** of the projection matrix!

Camera parameters: Summary

$$\mathbf{P} = \mathbf{K}[\mathbf{R}|\mathbf{t}]$$

- Intrinsic parameters
 - Principal point coordinates
 - Focal length
 - Pixel magnification factors
 - *Skew (non-rectangular pixels) – not important in practice*
 - *Radial distortion – important in practice!*

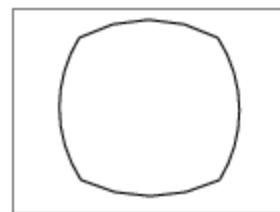
$$\mathbf{K} = \begin{bmatrix} m_x & 0 & 0 \\ 0 & m_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & \beta_x \\ 0 & \alpha_y & \beta_y \\ 0 & 0 & 1 \end{bmatrix}$$



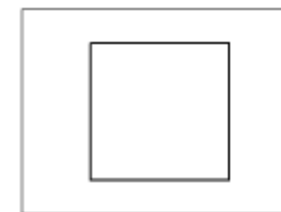
radial distortion



linear image



correction



camera calibration (estimation)

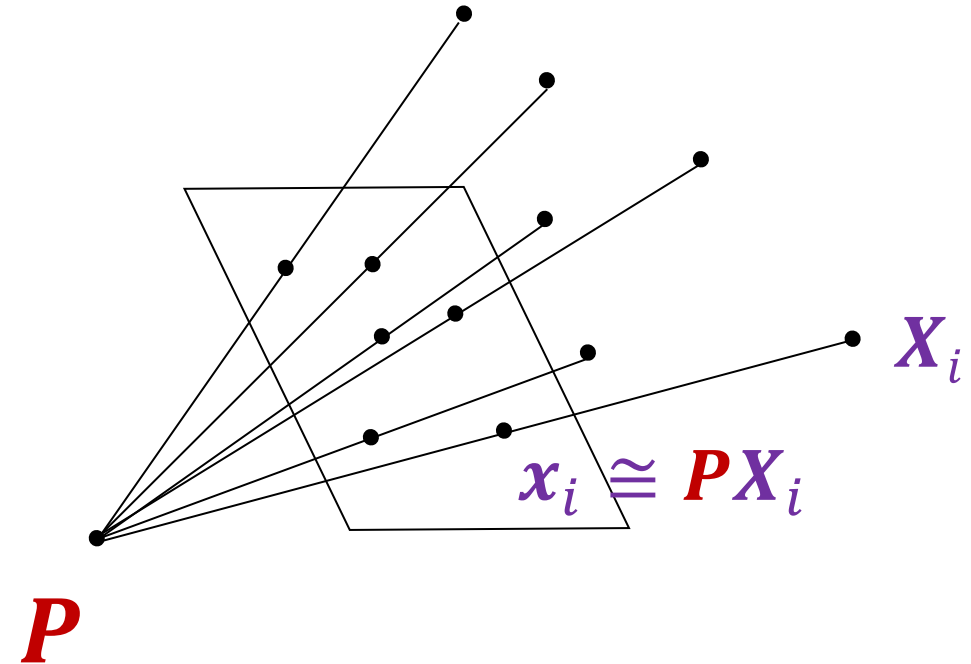
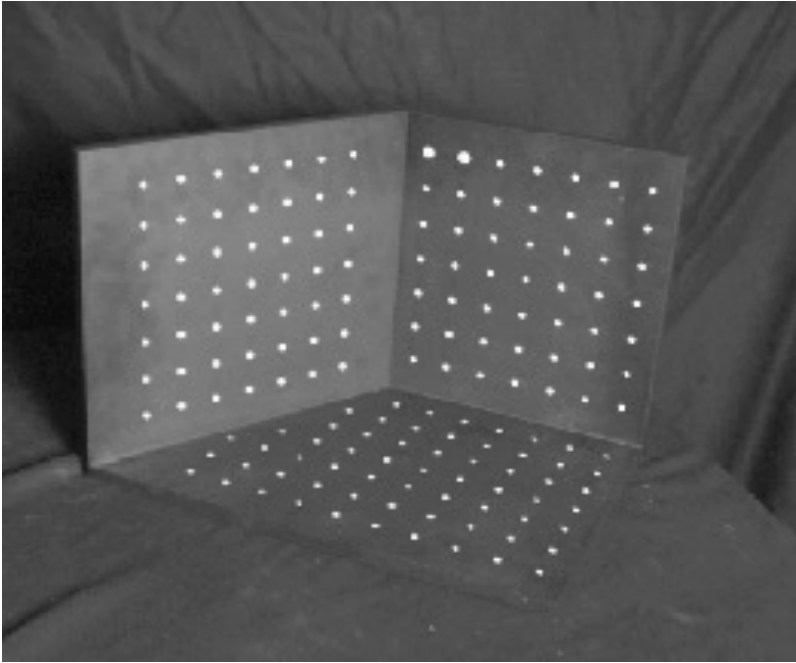
Camera calibration

$$\boldsymbol{x} \cong \boldsymbol{K}[\boldsymbol{R} \quad \boldsymbol{t}]\boldsymbol{X}$$

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \cong \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Camera calibration

- Given n points with known 3D coordinates X_i and known image projections x_i , estimate the camera parameters



Camera calibration

- Given n points with known 3D coordinates X_i and known image projections x_i , estimate the camera parameters



Known 2D image coords	Known 3D locations
--------------------------	-----------------------

880 214	312.747 309.140 30.086
43 203	305.796 311.649 30.356
270 197	307.694 312.358 30.418
886 347	310.149 307.186 29.298
745 302	311.937 310.105 29.216
943 128	311.202 307.572 30.682
476 590	307.106 306.876 28.660
419 214	309.317 312.490 30.230
317 335	307.435 310.151 29.318
783 521	308.253 306.300 28.881
235 427	306.650 309.301 28.905
665 429	308.069 306.831 29.189
655 362	309.671 308.834 29.029
427 333	308.255 309.955 29.267
412 415	307.546 308.613 28.963
746 351	311.036 309.206 28.913
434 415	307.518 308.175 29.069
525 234	309.950 311.262 29.990
716 308	312.160 310.772 29.080
602 187	311.988 312.709 30.514

Camera calibration: Linear method

- Recall homography fitting:

$$\mathbf{x}_i \cong \mathbf{P} \mathbf{X}_i \quad \mathbf{x}_i \times \mathbf{P} \mathbf{X}_i = \mathbf{0} \quad \begin{pmatrix} x_i \\ y_i \\ 1 \end{pmatrix} \times \begin{pmatrix} \mathbf{P}_1^T \mathbf{X}_i \\ \mathbf{P}_2^T \mathbf{X}_i \\ \mathbf{P}_3^T \mathbf{X}_i \end{pmatrix} = \mathbf{0}$$

$$\begin{bmatrix} \mathbf{0}^T & -\mathbf{X}_i^T & y_i \mathbf{X}_i^T \\ \mathbf{X}_i^T & \mathbf{0}^T & -x_i \mathbf{X}_i^T \\ -y_i \mathbf{X}_i^T & x_i \mathbf{X}_i^T & \mathbf{0}^T \end{bmatrix} \begin{pmatrix} \mathbf{P}_1 \\ \mathbf{P}_2 \\ \mathbf{P}_3 \end{pmatrix} = \mathbf{0}$$

One match gives **two** linearly independent constraints

Camera calibration: Linear method

- Final linear system:

$$\begin{bmatrix} \mathbf{0}^T & \mathbf{X}_1^T & -y_1 \mathbf{X}_1^T \\ \mathbf{X}_1^T & \mathbf{0}^T & -x_1 \mathbf{X}_1^T \\ \dots & \dots & \dots \\ \mathbf{0}^T & \mathbf{X}_n^T & -y_n \mathbf{X}_n^T \\ \mathbf{X}_n^T & \mathbf{0}^T & -x_n \mathbf{X}_n^T \end{bmatrix} \begin{pmatrix} \mathbf{P}_1 \\ \mathbf{P}_2 \\ \mathbf{P}_3 \end{pmatrix} = \mathbf{0} \quad \mathbf{A}\mathbf{p} = \mathbf{0}$$

- One 2D/3D correspondence gives two linearly independent equations
 - The projection matrix has 11 degrees of freedom
 - 6 correspondences needed for a minimal solution
- Homogeneous least squares: find $\mathbf{p}$ minimizing $\|\mathbf{A}\mathbf{p}\|^2$
 - Solution is eigenvector of $\mathbf{A}^T \mathbf{A}$ corresponding to smallest eigenvalue

Camera calibration: Linear method

- Final linear system:

$$\begin{bmatrix} \mathbf{0}^T & \mathbf{X}_1^T & -y_1 \mathbf{X}_1^T \\ \mathbf{X}_1^T & \mathbf{0}^T & -x_1 \mathbf{X}_1^T \\ \dots & \dots & \dots \\ \mathbf{0}^T & \mathbf{X}_n^T & -y_n \mathbf{X}_n^T \\ \mathbf{X}_n^T & \mathbf{0}^T & -x_n \mathbf{X}_n^T \end{bmatrix} \begin{pmatrix} \mathbf{P}_1 \\ \mathbf{P}_2 \\ \mathbf{P}_3 \end{pmatrix} = \mathbf{0} \quad \mathbf{A} \mathbf{p} = \mathbf{0}$$

- What if all the n 3D points are *coplanar*, i.e., there exists a set of plane parameters $\mathbf{\Pi}^T = (a, b, c, d)^T$ such that $\mathbf{\Pi}^T \mathbf{X}_i = 0$ for all i ?
 - Then we will get *degenerate solutions* $(\mathbf{\Pi}, \mathbf{0}, \mathbf{0})$, $(\mathbf{0}, \mathbf{\Pi}, \mathbf{0})$, or $(\mathbf{0}, \mathbf{0}, \mathbf{\Pi})$

Camera calibration: Linear vs. nonlinear

- Linear calibration is easy to formulate and solve, but it doesn't directly tell us the camera parameters

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \cong \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} \quad \text{vs.} \quad \boldsymbol{x} \cong \boldsymbol{K}[\boldsymbol{R} \mid \boldsymbol{t}]\boldsymbol{X}$$

Camera calibration: Linear vs. nonlinear

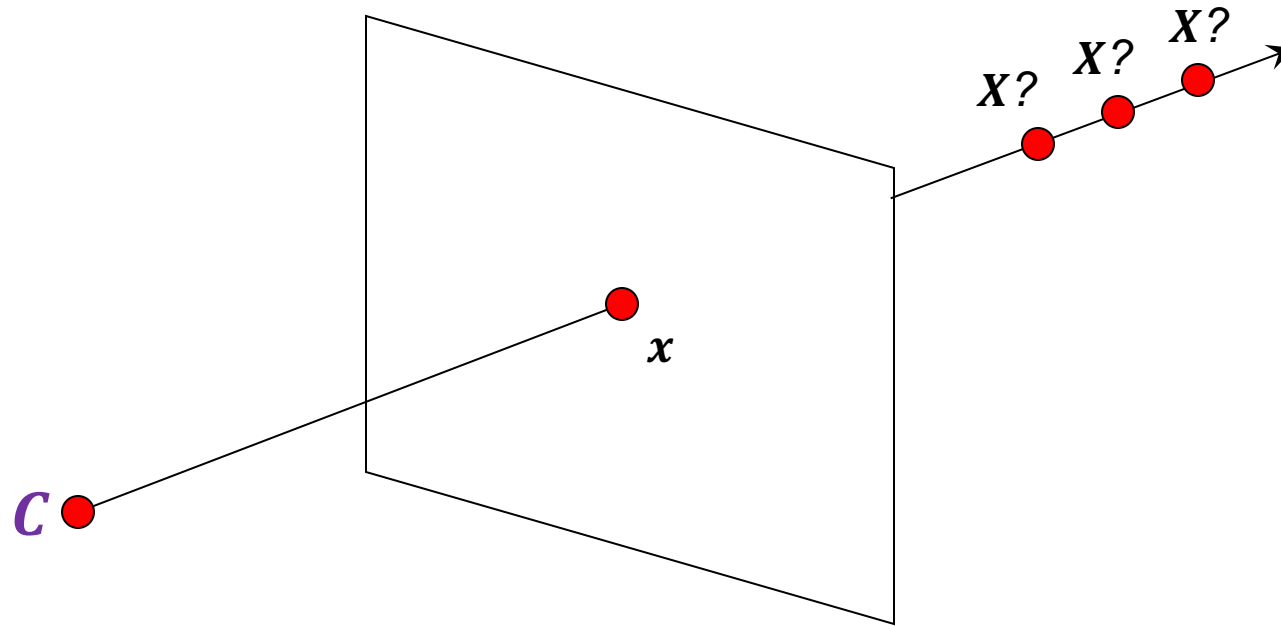
- Linear calibration is easy to formulate and solve, but it doesn't directly tell us the camera parameters
- In practice, *non-linear* methods are preferred
 - Write down objective function in terms of intrinsic and extrinsic parameters, as sum of squared distances between measured 2D points and estimated projections of 3D points:

$$\sum_i \|\text{proj}(\mathbf{K}[\mathbf{R} \mid \mathbf{t}]\mathbf{X}_i) - \mathbf{x}_i\|_2^2$$

- Can include radial distortion and constraints such as known focal length, orthogonality, visibility of points
- Minimize error using non-linear optimization package
- Can initialize solution with output of linear method (perform QR decomposition to get $\mathbf{K}$ and $\mathbf{R}$ from $\mathbf{P}$)

From image points to rays in space

- Calibration gives us a way to map image points to corresponding visual rays in the world
- We can define the visual ray using the camera center $\mathbf{C}$ (null space of $\mathbf{P}$) and any point $\mathbf{X}$ such that $\mathbf{x} \cong \mathbf{P}\mathbf{X}$



From image points to rays in space

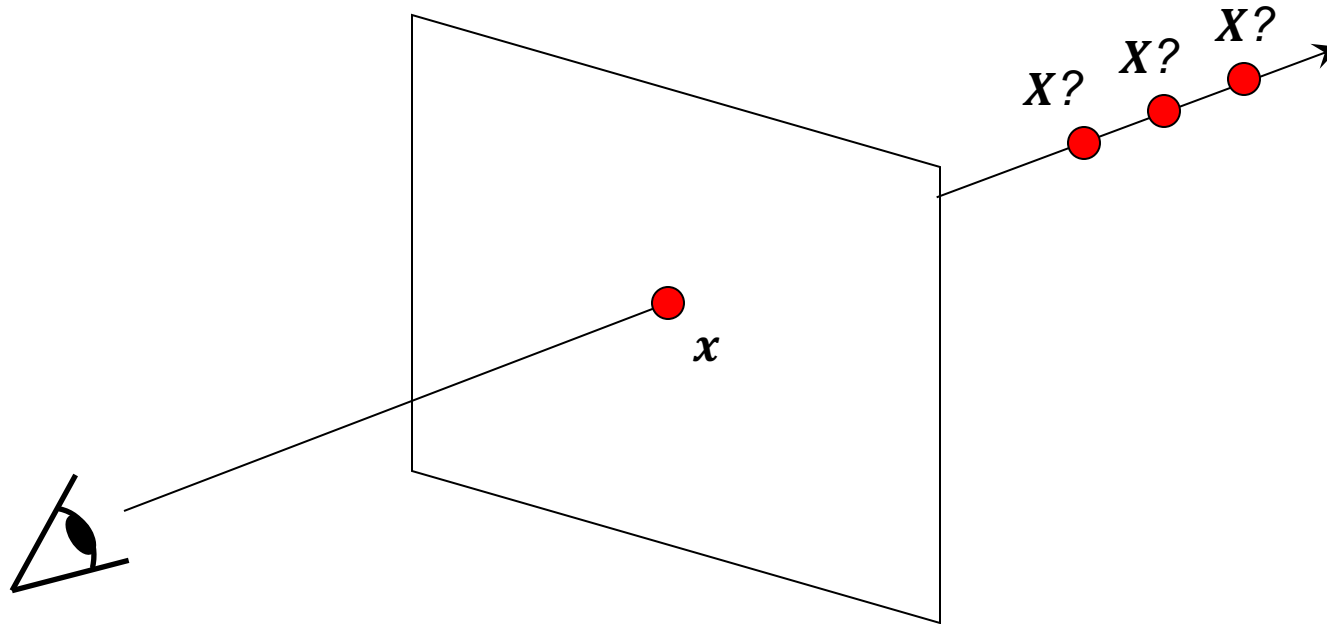
- Calibration gives us a way to map image points to corresponding visual rays in the world
- We can define the visual ray using the camera center $\mathbf{C}$ (null space of $\mathbf{P}$) and any point $\mathbf{X}$ such that $\mathbf{x} \cong \mathbf{PX}$
- In particular, we can let $\mathbf{X} = \mathbf{P}^+ \mathbf{x}$ where $\mathbf{P}^+ = \mathbf{P}^T (\mathbf{P} \mathbf{P}^T)^{-1}$ is the *pseudoinverse* of the camera projection matrix
 - It's a 4×3 matrix such that $\mathbf{P} \mathbf{P}^+ = \mathbf{I}$
 - If $\mathbf{X} = \mathbf{P}^+ \mathbf{x}$, then $\mathbf{PX} = \mathbf{P} \mathbf{P}^+ \mathbf{x} = \mathbf{x}$

first taste of 3D reconstruction

Our goal: Recovery of 3D structure



Single-view ambiguity



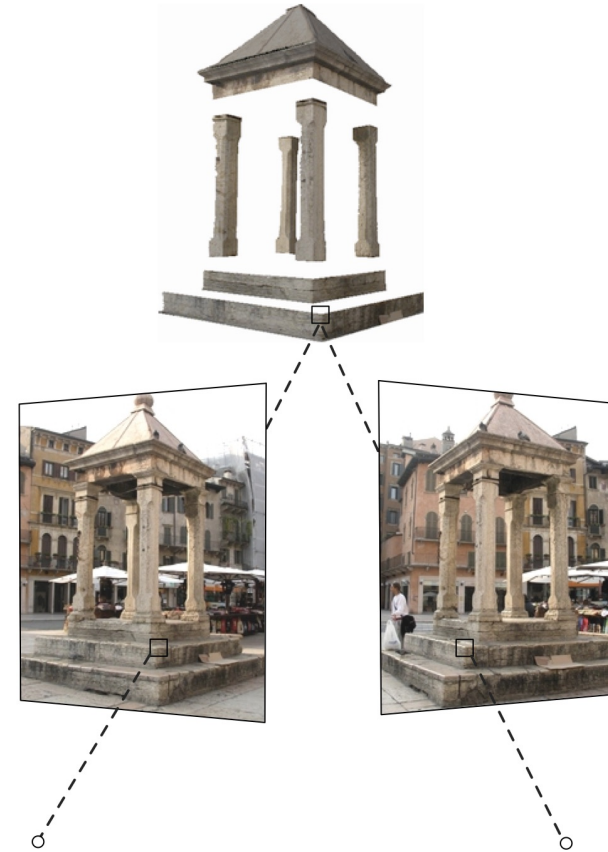
Single-view ambiguity



[Rashad Alakbarov shadow sculptures](#)

Our goal: Recovery of 3D structure

- When certain assumptions hold, we can recover structure from a single view
- In general, we need *multi-view geometry*



[Image source](#)

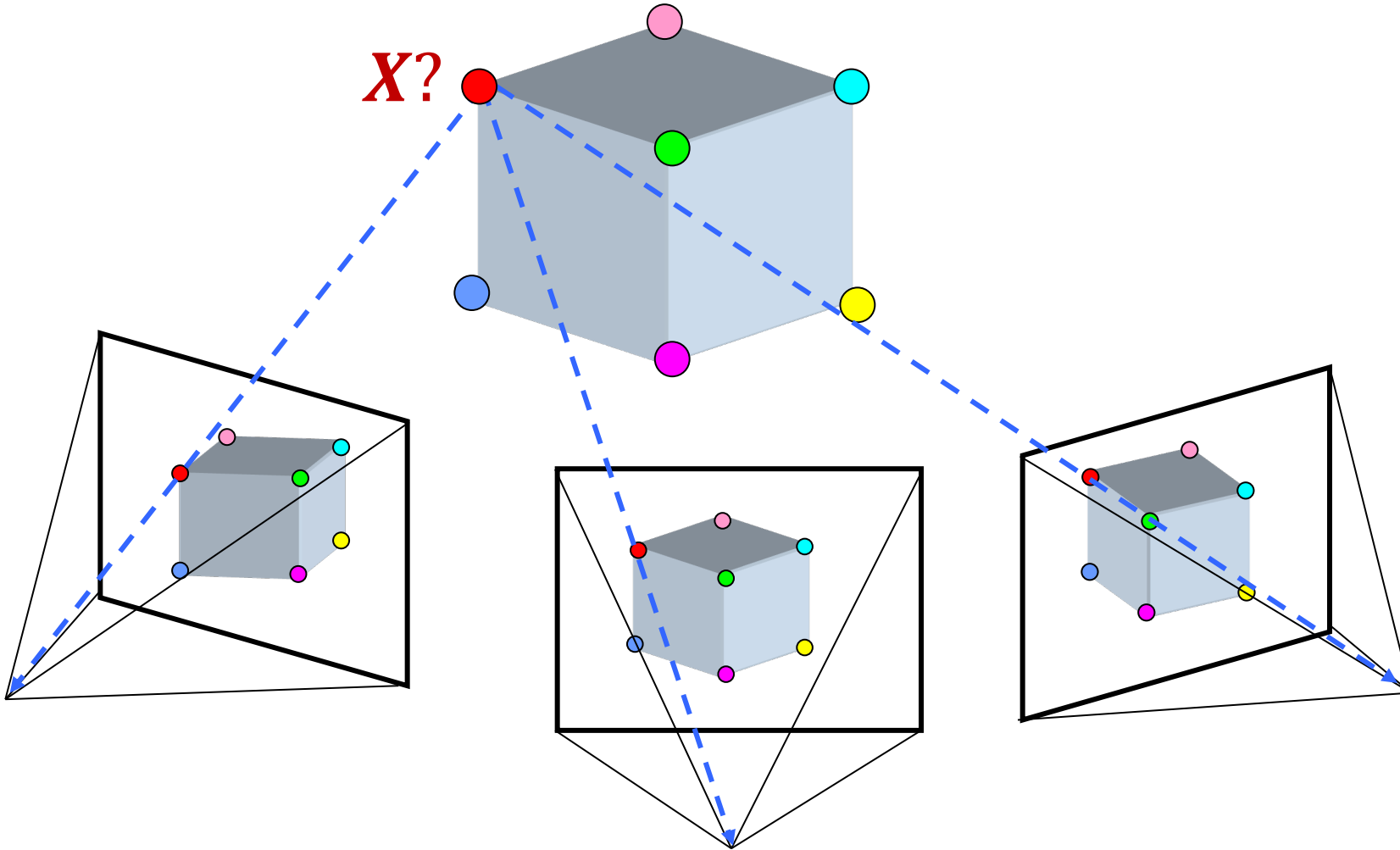
Triangulation

- Given projections of a 3D point in two or more images (with known camera matrices), find the coordinates of the point



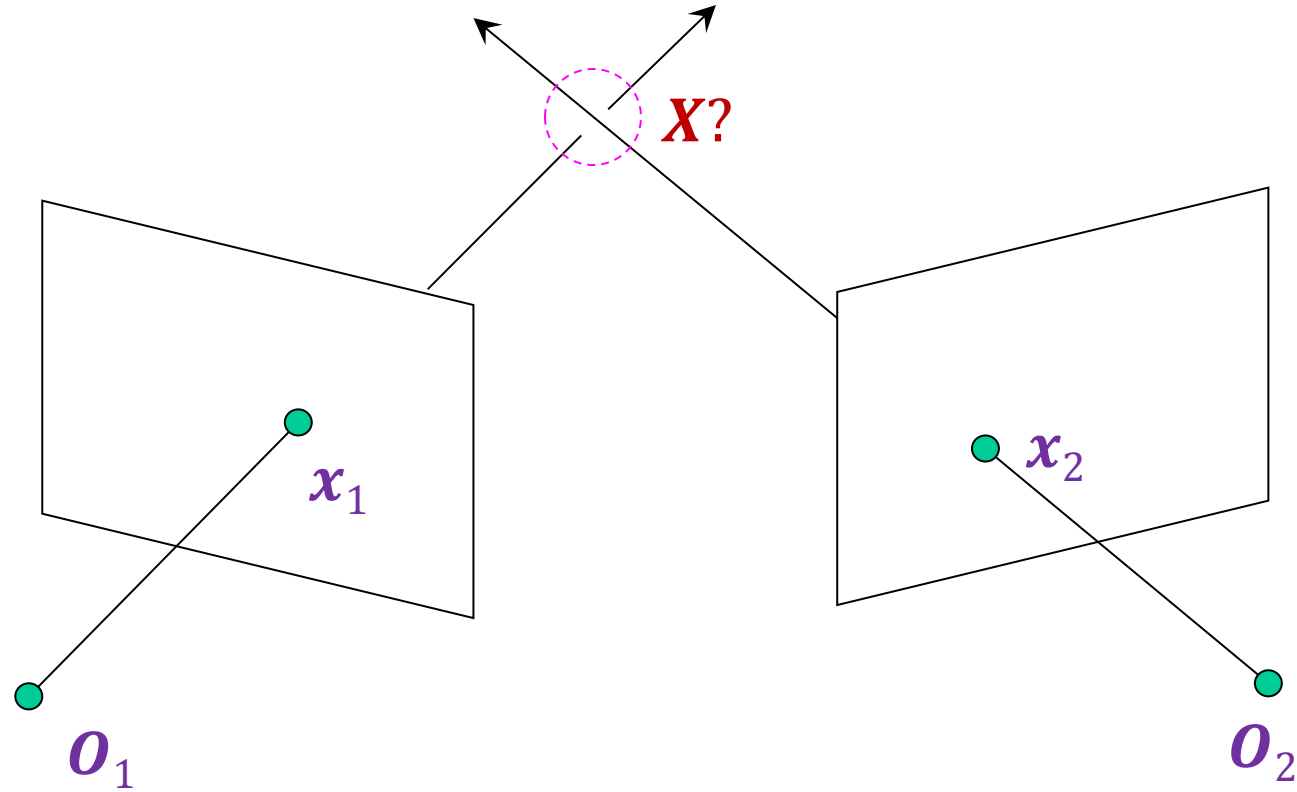
Triangulation

- Given projections of a 3D point in two or more images (with known camera matrices), find the coordinates of the point



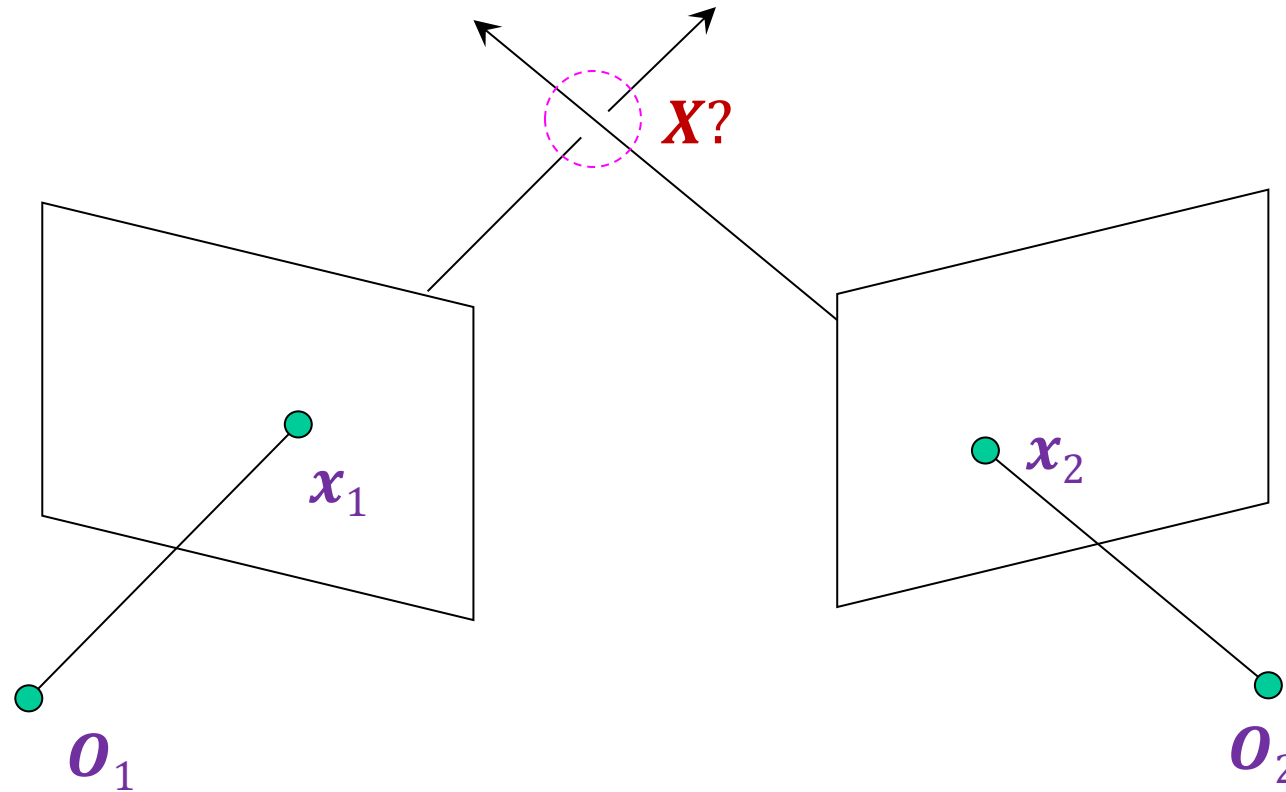
Triangulation

- Given projections of a 3D point in two or more images (with known camera matrices), find the coordinates of the point



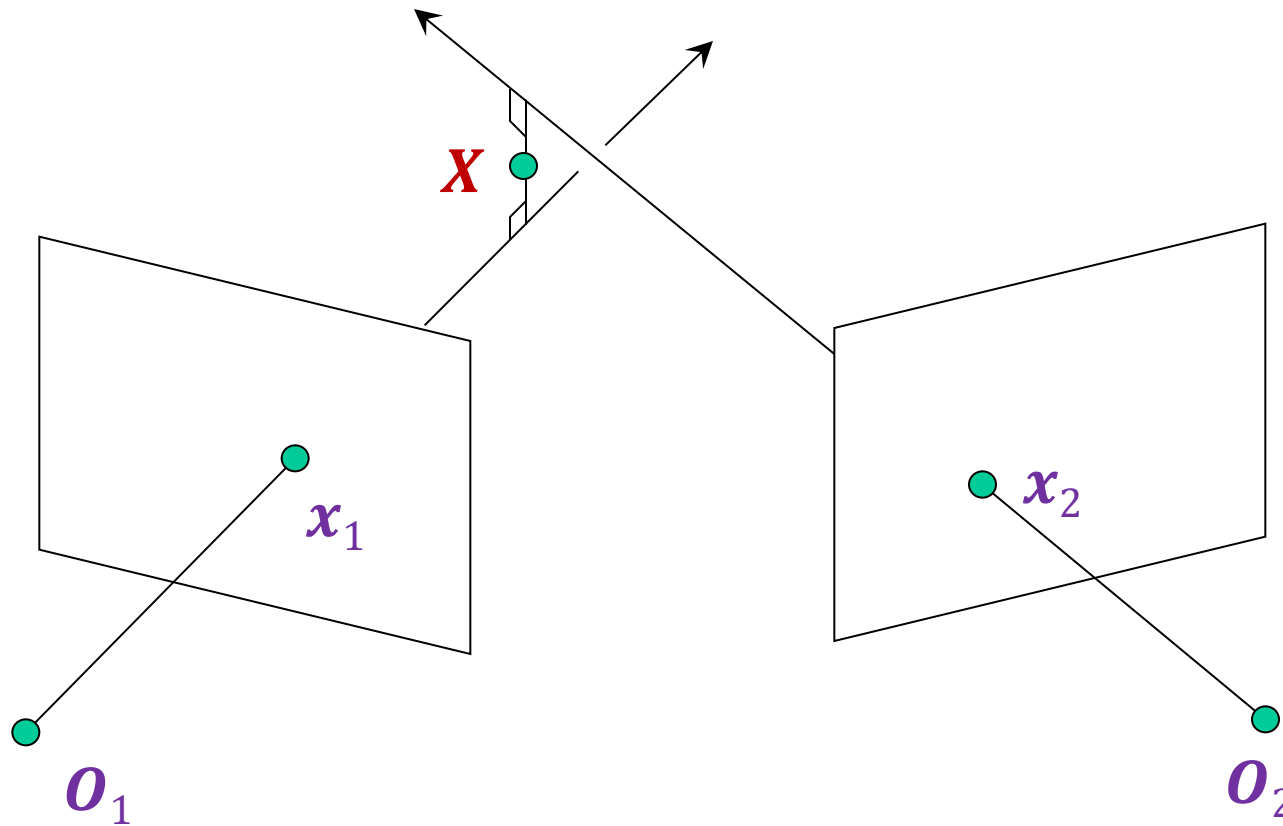
Triangulation

- We want to intersect the two visual rays corresponding to x_1 and x_2 , but because of noise and numerical errors, they don't meet exactly



Triangulation: Geometric approach

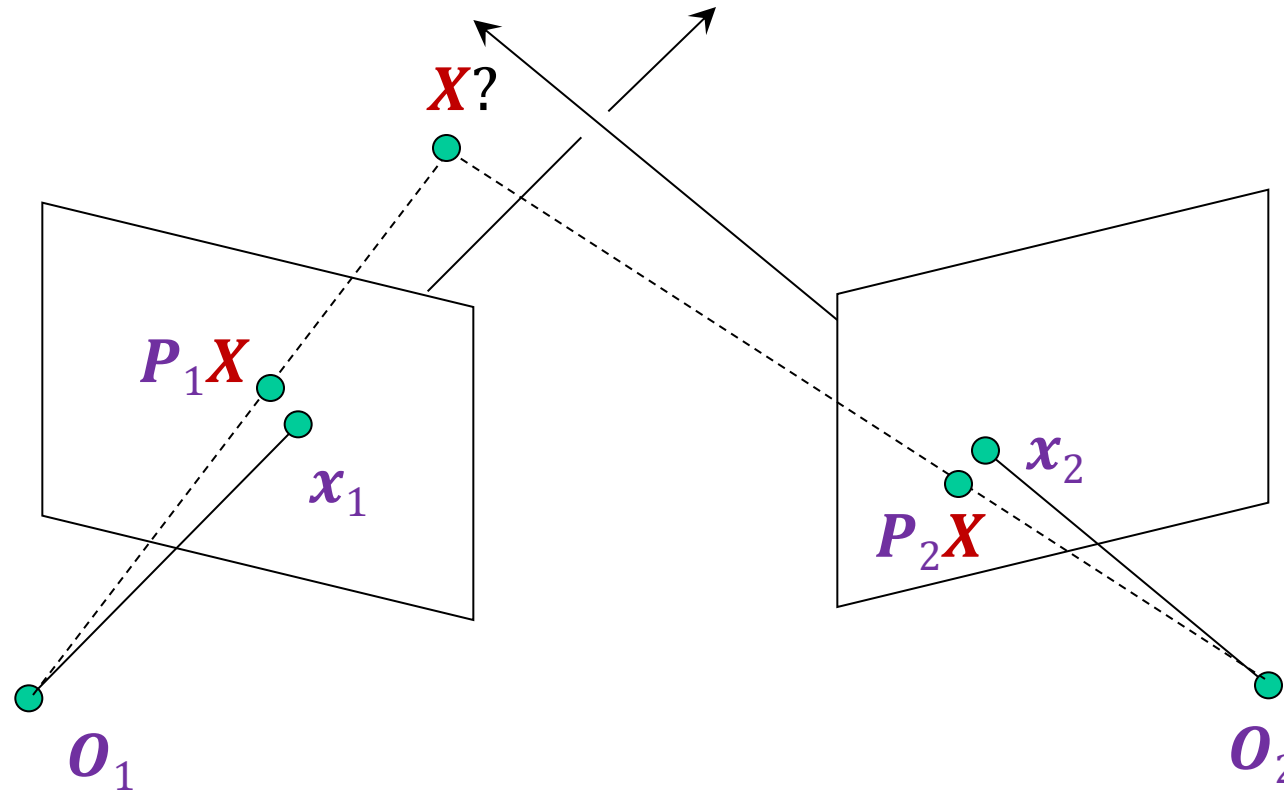
- Find shortest segment connecting the two viewing rays and let X be the midpoint of that segment



Triangulation: Nonlinear approach

- Find $\mathbf{X}$ that minimizes

$$\|\text{proj}(\mathbf{P}_1 \mathbf{X}) - \mathbf{x}_1\|_2^2 + \|\text{proj}(\mathbf{P}_2 \mathbf{X}) - \mathbf{x}_2\|_2^2$$



Triangulation: Linear approach

$$\mathbf{x}_1 \cong \mathbf{P}_1 \mathbf{X}$$

$$\mathbf{x}_2 \cong \mathbf{P}_2 \mathbf{X}$$

$$\mathbf{x}_1 \times \mathbf{P}_1 \mathbf{X} = \mathbf{0}$$

$$\mathbf{x}_2 \times \mathbf{P}_2 \mathbf{X} = \mathbf{0}$$

$$[\mathbf{x}_{1 \times}] \mathbf{P}_1 \mathbf{X} = \mathbf{0}$$

$$[\mathbf{x}_{2 \times}] \mathbf{P}_2 \mathbf{X} = \mathbf{0}$$

- Rewrite cross product as matrix multiplication:

$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = [\mathbf{a}_{\times}] \mathbf{b}$$

Triangulation: Linear approach

$$\mathbf{x}_1 \cong \mathbf{P}_1 \mathbf{X}$$

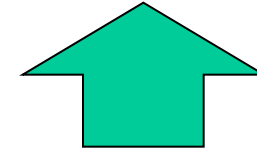
$$\mathbf{x}_2 \cong \mathbf{P}_2 \mathbf{X}$$

$$\mathbf{x}_1 \times \mathbf{P}_1 \mathbf{X} = \mathbf{0}$$

$$\mathbf{x}_2 \times \mathbf{P}_2 \mathbf{X} = \mathbf{0}$$

$$[\mathbf{x}_1 \times] \mathbf{P}_1 \mathbf{X} = \mathbf{0}$$

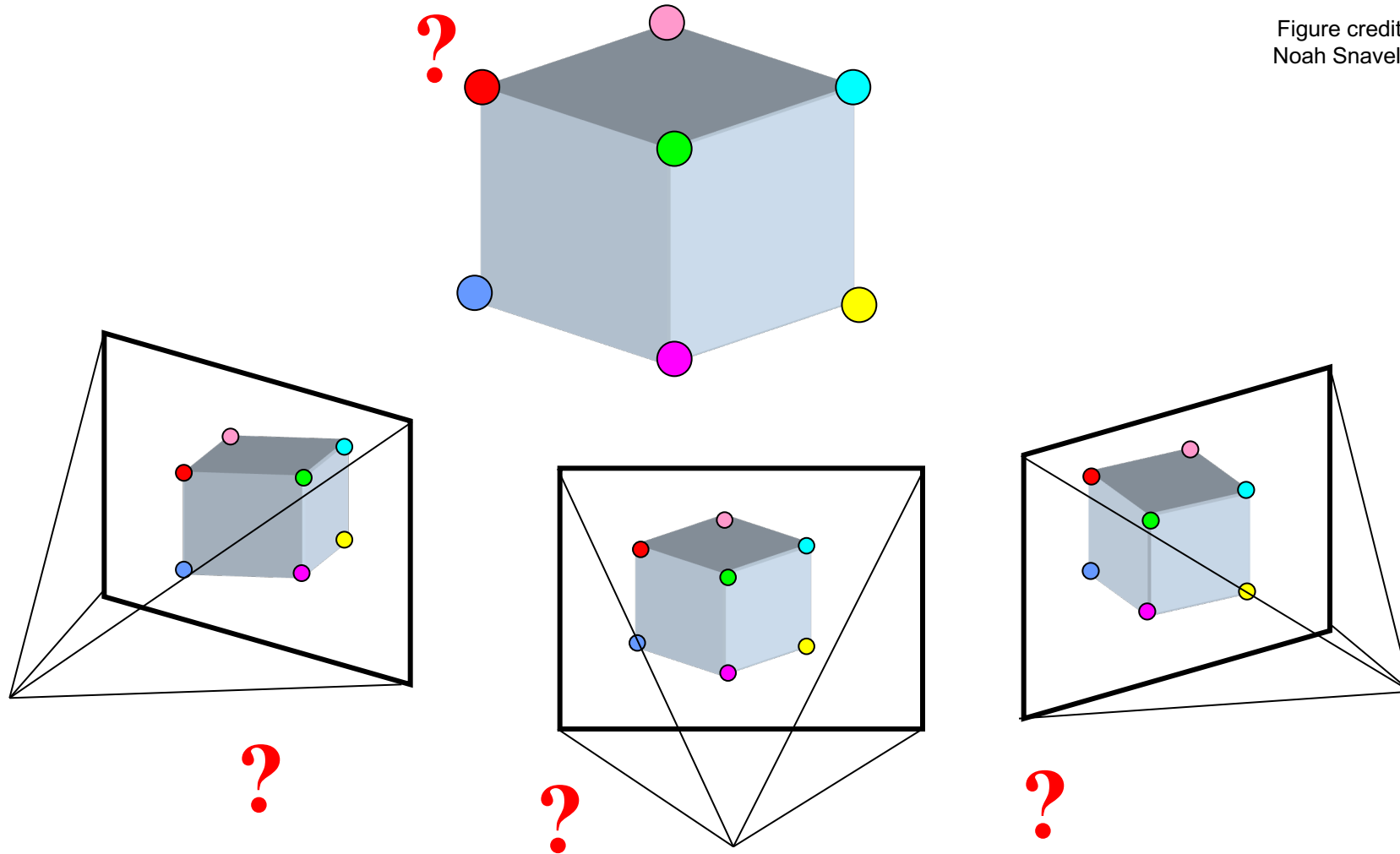
$$[\mathbf{x}_2 \times] \mathbf{P}_2 \mathbf{X} = \mathbf{0}$$



Two independent
equations each in terms of
three unknown entries of $\mathbf{X}$

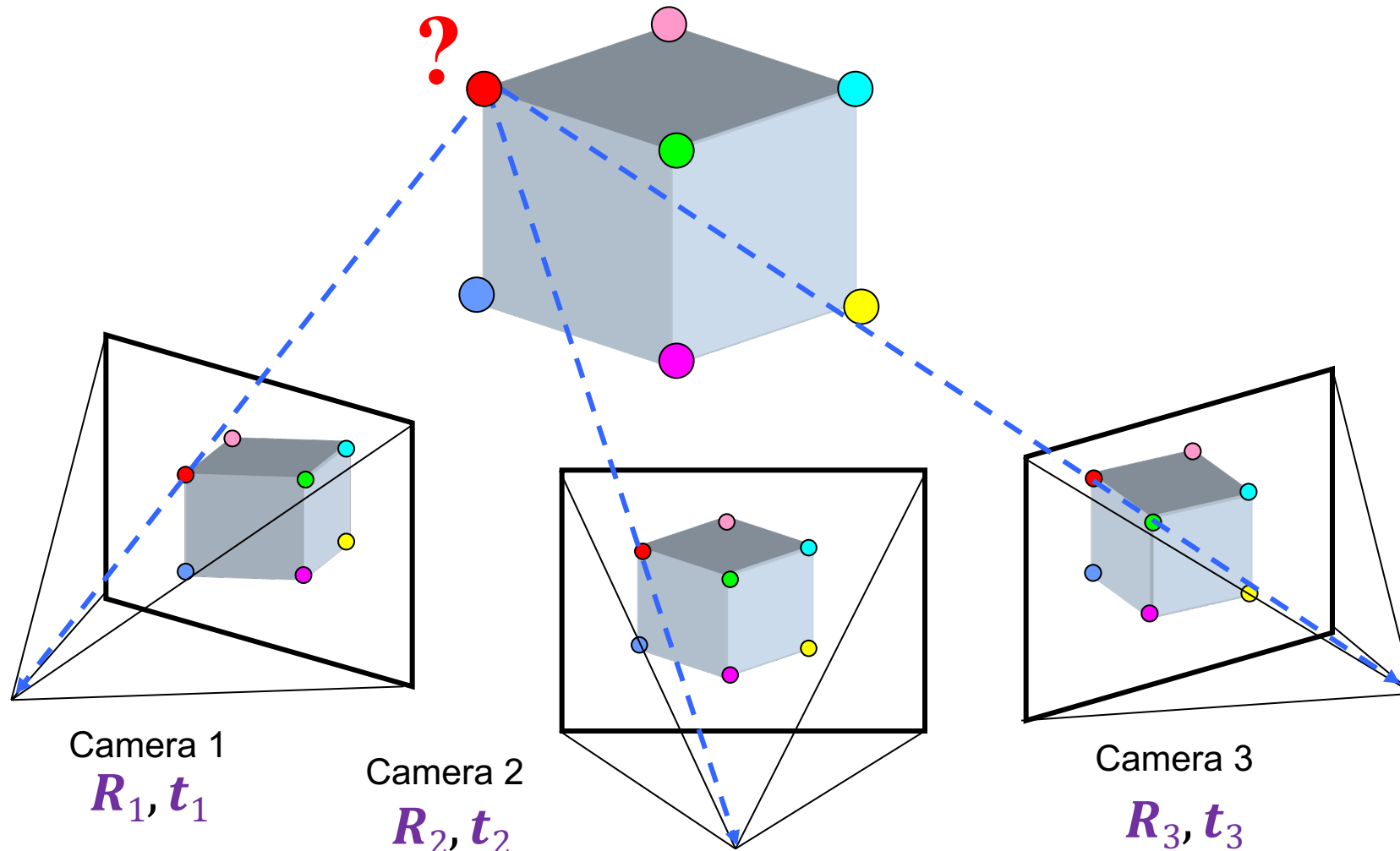
Preview: Structure from motion

Figure credit:
Noah Snavely



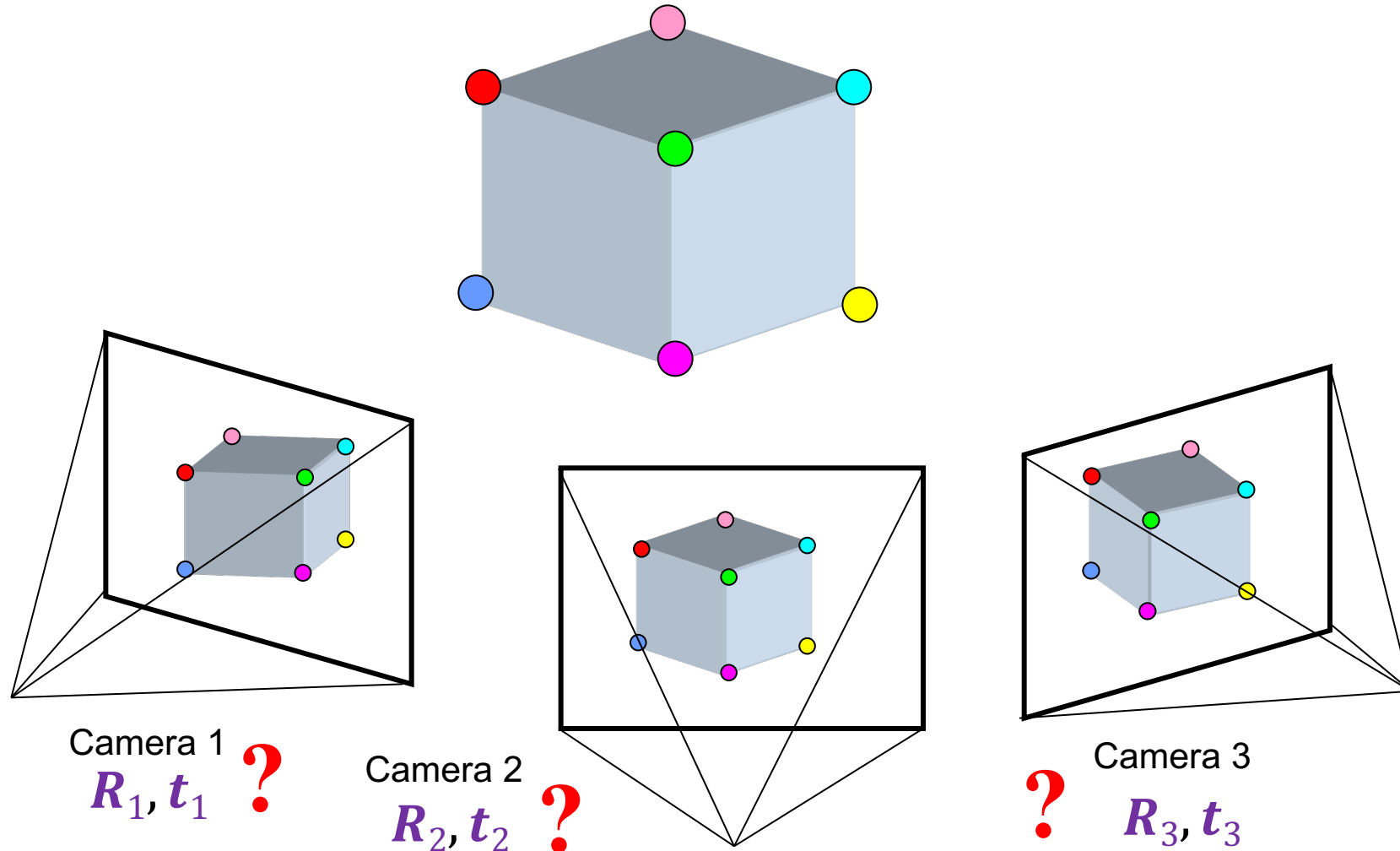
- Given 2D point correspondences between multiple images, compute the camera parameters and the 3D points

Preview: Structure from motion



- **Structure:** Given *known cameras* and projections of the same 3D point in two or more images, compute the 3D coordinates of that point
 - Triangulation!

Preview: Structure from motion



- **Motion:** Given a set of *known* 3D points seen by a camera, compute the camera parameters
 - Calibration!

Multi-view geometry “Bible”

